

Koran

43 topics · 111 questions. Each day 10 are picked at random. True/false has one try; other questions have two. Any topic missed comes back the next day with a different question. Correct answers are in green (the dashboard shuffles the answer order).

Anatomy & physiology 6 topics

Anatomy vs. physiology

1. MULTIPLE CHOICE

Anatomy is the study of:

✓ **Body structure and the physical relationships between body parts**

How the body functions

Only diseases of the body

Only the cells of the body

2. TRUE OR FALSE

Physiology is the study of how the body functions.

Answer: **True**

3. ANALYZE THE CASE

A student measures how fast the heart pumps blood during exercise. Is the student studying anatomy or physiology?

✓ **Physiology**

Anatomy

Cytology

Surface anatomy

Explanation shown if missed: Anatomy = STRUCTURE (what a body part is and where it is). Physiology = FUNCTION (how it works). Studying the shape of the heart is anatomy; studying how the heart pumps is physiology.

Gross vs. microscopic anatomy

1. MULTIPLE CHOICE

Which area of anatomy examines structures visible to the naked eye?

✓ **Gross (macroscopic) anatomy**

Microscopic anatomy

Cytology

Histology

2. ANALYZE THE CASE

A scientist studies a thin slice of the stomach wall under a microscope. Which area of anatomy is this?

✓ **Microscopic anatomy**

Gross anatomy

Surface anatomy

Regional anatomy

3. TRUE OR FALSE

Microscopic anatomy studies structures that are visible to the naked eye.

Answer: **False**

Explanation shown if missed: Gross (macroscopic) anatomy looks at things you can see with your eyes, like the heart or a bone. Microscopic anatomy looks at things too small to see without a microscope.

Cytology & histology

1. MULTIPLE CHOICE

Histology is the study of:

✓ **Tissues**

Cells

Organ systems

Body regions

2. TRUE OR FALSE

Cytology is the study of cells.

Answer: **True**

3. ANALYZE THE CASE

A lab looks at individual cells scraped from the inside of the cheek. Which field is this?

✓ **Cytology**

Histology

Regional anatomy

Systemic physiology

Explanation shown if missed: Cytology = the study of CELLS (think "cyto" = cell). Histology = the study of TISSUES, which are groups of cells working together.

Surface, regional & systemic anatomy

1. MULTIPLE CHOICE

Which area of anatomy studies general form and superficial markings?

✓ **Surface anatomy**

Regional anatomy

Microscopic anatomy

Developmental anatomy

2. ANALYZE THE CASE

A textbook chapter covers every structure in the head and neck together. Which area of anatomy is it using?

✓ **Regional anatomy**

Systemic anatomy

Cytology

Developmental anatomy

3. ANALYZE THE CASE

A class studies the whole digestive system, from mouth to rectum. Which area of anatomy is this?

✓ **Systemic anatomy**

Regional anatomy

Surface anatomy

Histology

Explanation shown if missed: Surface anatomy = outside form and markings. Regional anatomy = everything in one body AREA (like the head and neck). Systemic anatomy = one organ SYSTEM at a time (like the digestive system).

Developmental & clinical anatomy

1. MULTIPLE CHOICE

Which area of anatomy studies changes in form from conception through physical maturity?

✓ **Developmental anatomy**

Clinical anatomy

Surface anatomy

Systemic anatomy

2. ANALYZE THE CASE

A doctor uses knowledge of anatomy to find the cause of a patient's pain and plan treatment. Which area is this?

✓ **Clinical (pathological) anatomy**

Developmental anatomy

Cytology

Surface anatomy

Explanation shown if missed: Developmental anatomy follows how the body's form changes from conception until physical maturity. Clinical (pathological) anatomy uses anatomy to diagnose and treat patients.

Areas of physiology

1. **MULTIPLE CHOICE**

Which area of physiology studies chemical processes within and between cells?

✓ **Cell physiology**

Systemic physiology

Special (organ) physiology

Clinical physiology

2. **ANALYZE THE CASE**

A researcher studies how diabetes damages the kidneys. Which area of physiology is this?

✓ **Clinical (pathological) physiology**

Cell physiology

Special (organ) physiology

Systemic physiology

3. **TRUE OR FALSE**

Special (organ) physiology studies the functions of specific organs.

Answer: **True**

Explanation shown if missed: Cell physiology = chemical processes in and between cells. Special (organ) physiology = one specific organ. Systemic physiology = a whole organ system. Clinical (pathological) physiology = how disease affects organs or systems.

Organization & homeostasis 8 topics

Levels of organization

1. **MULTIPLE CHOICE**

Which list shows the levels of organization in the correct order, from smallest to largest?

✓ **Atoms → Molecules → Organelles → Cells → Tissues → Organs → Organ systems → Organism**

Cells → Atoms → Molecules → Tissues → Organelles → Organs → Organ systems → Organism

Molecules → Atoms → Cells → Organelles → Organs → Tissues → Organism → Organ systems

Atoms → Molecules → Cells → Organelles → Tissues → Organ systems → Organs → Organism

2. **MULTIPLE CHOICE**

Which level comes right after TISSUES?

✓ **Organs**

Cells

Organ systems

Organelles

3. **TRUE OR FALSE**

An organ is composed of tissues.

Answer: **True**

Explanation shown if missed: From smallest to largest: Atoms → Molecules → Organelles → Cells → Tissues → Organs → Organ systems → Organism. Small pieces build bigger pieces, like letters → words → sentences → a story.

Atom vs. cell

1. **MULTIPLE CHOICE**

What is the smallest LIVING unit of the body?

✓ **Cell**

Atom

Molecule

Organelle

2. **TRUE OR FALSE**

The atom is the smallest living unit of the body.

Answer: **False**

Explanation shown if missed: The atom is the smallest stable unit of matter, but it is NOT alive. The cell is the smallest LIVING unit.

Organ system

1. **MULTIPLE CHOICE**

An organ system is:

✓ **A group of organs with a unique collective function**

A group of similar cells

The smallest living unit

A single organ

2. ANALYZE THE CASE

The mouth, stomach, and intestines work together to break down food. This group is an example of a(n):

✓ **Organ system**

Tissue

Organelle

Molecule

Explanation shown if missed: An organ system is a group of organs that work together for one shared job. Example: the heart, blood vessels, and blood work together to move blood around the body.

Characteristics of life

1. SELECT ALL

Select the characteristics of life listed in the review guide.

✓ **Cells**

✓ **Metabolism**

✓ **Reproduction**

✓ **Response to the environment**

Photosynthesis

Living only in water

2. MULTIPLE CHOICE

In the memory trick CHRMD, what does the M stand for?

✓ **Metabolism**

Movement

Muscles

Membranes

3. ANALYZE THE CASE

You touch a hot pan and pull your hand away. Which characteristic of life does this show?

✓ **Response to the environment**

Reproduction

Adaptation over generations

Heredity

Explanation shown if missed: Living things have cells, metabolism, reproduction, response to the environment, and adaptation (evolution). The slides' memory trick is CHRMD: Cells, Homeostasis, Reproduction, Metabolism, DNA/Heredity.

Homeostasis

1. MULTIPLE CHOICE

Homeostasis is:

✓ **Maintenance of a relatively stable internal environment**

The production of new cells

The study of tissues

An increase in body size

2. ANALYZE THE CASE

On a hot day you start to sweat and your body temperature stays near normal.

What is your body doing?

✓ **Maintaining homeostasis**

Using positive feedback

Reproducing

Changing its anatomy

3. TRUE OR FALSE

Homeostasis means the internal environment changes wildly all the time.

Answer: **False**

Explanation shown if missed: Homeostasis means keeping the inside of your body relatively stable, even when things outside change. Example: sweating on a hot day keeps your body temperature near normal.

Negative feedback

1. MULTIPLE CHOICE

In negative feedback, the response:

✓ **Opposes/reduces the original stimulus**

Enhances/amplifies the original stimulus

Has no effect on the stimulus

Stops all body functions

2. ANALYZE THE CASE

After a meal your blood glucose rises; insulin is released and glucose drops back to normal. This is an example of:

✓ **Negative feedback**

Positive feedback

Cardiac tamponade

Developmental anatomy

3. TRUE OR FALSE

Control of body temperature is an example of negative feedback.

Answer: **True**

Explanation shown if missed: In negative feedback, the body's response goes the OPPOSITE way of the change, bringing things back to normal. Examples: body temperature, blood pressure, blood glucose, and blood calcium.

Positive feedback

1. MULTIPLE CHOICE

In positive feedback, the response:

✓ **Enhances/amplifies the original stimulus**

Opposes/reduces the original stimulus

Always restores the normal set point right away

Only happens in plants

2. ANALYZE THE CASE

During childbirth, contractions trigger a hormone that makes the contractions even stronger. This is:

✓ **Positive feedback**

Negative feedback

Homeostatic set point

A sectional plane

3. SELECT ALL

Select the examples of POSITIVE feedback.

✓ **Childbirth**

✓ **Blood clotting**

✓ **Lactation**

Body temperature regulation

Blood pressure regulation

Explanation shown if missed: In positive feedback, the response makes the change even BIGGER until an event is finished. Examples: childbirth (contractions get stronger), blood clotting, and lactation.

Receptor → control center → effector

1. MULTIPLE CHOICE

What is the correct order of a homeostatic regulatory mechanism?

✓ **Receptor → Control center → Effector**

Effector → Receptor → Control center

Control center → Effector → Receptor

Receptor → Effector → Control center

2. ANALYZE THE CASE

Skin sensors detect cold, the brain processes it, and muscles start to shiver. In this example, the muscles are the:

✓ **Effector**

Receptor

Control center

Stimulus

3. TRUE OR FALSE

The control center is the part that senses a change.

Answer: **False**

Explanation shown if missed: Receptor = senses a change. Control center = processes the information and sends instructions. Effector = carries out the instructions. Example: skin senses cold → brain decides → muscles shiver.

Position, directions & planes 11 topics

Anatomical position

1. MULTIPLE CHOICE

Which describes the anatomical position?

✓ **Standing erect, facing forward, arms at the sides, palms forward, feet slightly apart**

Lying face down with arms over the head

Standing with palms facing backward and feet crossed

Sitting with arms folded

2. TRUE OR FALSE

In anatomical position, the palms face backward.

Answer: **False**

Explanation shown if missed: Anatomical position: standing up straight, facing forward, arms at the sides, palms facing FORWARD, feet slightly apart. Doctors use it so everyone describes the body the same way.

Supine vs. prone

1. MULTIPLE CHOICE

A patient lying on their back with the face upward is in the ___ position.

✓ **Supine**

Prone

Anatomical

Lateral

2. ANALYZE THE CASE

For an exam of the back, the patient lies on their belly with the face down. This position is:

✓ **Prone**

Supine

Anatomical

Superior

3. TRUE OR FALSE

Prone means lying face down.

Answer: **True**

Explanation shown if missed: Supine = lying on your back, face UP (you could hold a bowl of soup: "sup"). Prone = lying on your belly, face DOWN.

Superior vs. inferior

1. MULTIPLE CHOICE

The head is ___ to the chest.

✓ **Superior**

Inferior

Distal

Deep

2. TRUE OR FALSE

Inferior means toward the feet.

Answer: **True**

3. ANALYZE THE CASE

The knee is closer to the feet than the hip is. The knee is ___ to the hip.

✓ **Inferior**

Superior

Medial

Superficial

Explanation shown if missed: Superior = toward the HEAD (higher). Inferior = toward the FEET (lower). The head is superior to the chest.

Anterior vs. posterior

1. MULTIPLE CHOICE

The sternum (breastbone) is ___ to the spine.

✓ **Anterior (ventral)**

Posterior (dorsal)

Lateral

Proximal

2. TRUE OR FALSE

Dorsal means the same as posterior.

Answer: **True**

3. ANALYZE THE CASE

The shoulder blades are on the back of the body and the belly button is on the front. The shoulder blades are ___ to the belly button.

✓ **Posterior (dorsal)**

Anterior (ventral)

Distal

Medial

Explanation shown if missed: Anterior (ventral) = toward the FRONT. Posterior (dorsal) = toward the BACK. The breastbone is anterior to the spine.

Medial vs. lateral

1. MULTIPLE CHOICE

The nose is ___ to the eyes.

✓ **Medial**

Lateral

Distal

Superficial

2. ANALYZE THE CASE

The ears are on the sides of the head, far from the midline. The ears are ___ to the nose.

✓ **Lateral**

Medial

Proximal

Deep

3. TRUE OR FALSE

Lateral means toward the midline of the body.

Answer: **False**

Explanation shown if missed: Medial = toward the MIDLINE (an imaginary line down the middle of the body). Lateral = AWAY from the midline, toward the sides.

Proximal vs. distal

1. MULTIPLE CHOICE

The elbow is ___ to the wrist.

✓ **Proximal**

Distal

Lateral

Inferior

2. ANALYZE THE CASE

**The fingers are farther than the elbow from where the arm attaches to the body.
The fingers are ___ to the elbow.**

✓ **Distal**

Proximal

Medial

Superior

Explanation shown if missed: Proximal = CLOSER to where a limb attaches to the body. Distal = FARTHER from that point. The elbow is proximal to the wrist; the fingers are distal to the elbow.

Superficial vs. deep

1. MULTIPLE CHOICE

The skin is ___ to the muscles.

✓ **Superficial**

Deep

Proximal

Caudal

2. TRUE OR FALSE

Deep means farther from the surface of the body.

Answer: **True**

Explanation shown if missed: Superficial = closer to the body SURFACE. Deep = farther INSIDE, away from the surface. The skin is superficial to the muscles.

Cranial vs. caudal

1. MULTIPLE CHOICE

Caudal means:

✓ **Toward the lower end or tail**

Toward the head

Toward the front

Toward the midline

2. TRUE OR FALSE

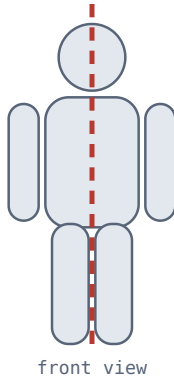
Cranial means toward the head.

Answer: **True**

Explanation shown if missed: Cranial = toward the HEAD (cranium = skull). Caudal = toward the lower end or TAIL ("cauda" means tail).

Sagittal plane

1. MULTIPLE CHOICE



Which plane is shown by the dashed line?

✓ **Sagittal**

Coronal (frontal)

Transverse (horizontal)

Diagonal

2. MULTIPLE CHOICE

Which plane divides the body into left and right portions?

✓ **Sagittal**

Coronal (frontal)

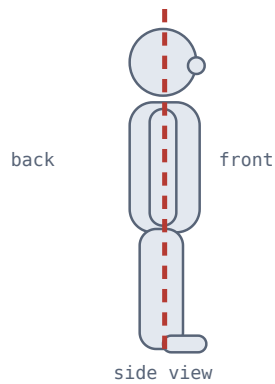
Transverse (horizontal)

Superficial

Explanation shown if missed: A sagittal plane cuts the body into LEFT and RIGHT portions. Memory trick: Sagittal = Sides.

Coronal (frontal) plane

1. MULTIPLE CHOICE



In this side view, which plane is shown by the dashed line?

✓ **Coronal (frontal)**

Sagittal

Transverse (horizontal)

Diagonal

2. ANALYZE THE CASE

A cut separates the face and chest from the back of the body. Which plane made this cut?

✓ **Coronal (frontal)**

Sagittal

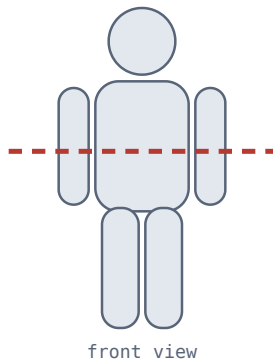
Transverse (horizontal)

Midline

Explanation shown if missed: A coronal (frontal) plane cuts the body into a FRONT (anterior) piece and a BACK (posterior) piece. Memory trick: Coronal = front/back.

Transverse (horizontal) plane

1. MULTIPLE CHOICE



Which plane is shown by the dashed line?

✓ **Transverse (horizontal)**

Sagittal

Coronal (frontal)

Vertical

2. ANALYZE THE CASE

A CT scan shows "slices" that separate the body into upper and lower parts. Which plane are these slices in?

✓ **Transverse (horizontal)**

Sagittal

Coronal (frontal)

Oblique

Explanation shown if missed: A transverse (horizontal) plane cuts the body into a TOP (superior) piece and a BOTTOM (inferior) piece. Memory trick: Transverse = top/bottom, like slicing bread.

Body cavities & membranes 6 topics

Ventral body cavity

1. MULTIPLE CHOICE

The ventral body cavity includes the:

✓ **Thoracic and abdominopelvic cavities**

Cranial and spinal cavities

Cranial cavity only

Pericardial and cranial cavities

2. TRUE OR FALSE

The diaphragm separates the thoracic cavity from the abdominopelvic cavity.

Answer: **True**

3. ANALYZE THE CASE

A wound tears through the diaphragm. Which two cavities are no longer separated?

✓ **Thoracic and abdominopelvic**

Cranial and spinal

Pericardial and pleural

Spinal and abdominopelvic

Explanation shown if missed: The ventral (front) cavity has two parts: the thoracic cavity (chest) and the abdominopelvic cavity (belly and pelvis). The diaphragm muscle separates them.

Dorsal body cavity

1. MULTIPLE CHOICE

The dorsal body cavity contains the:

✓ **Cranial cavity (brain) and spinal cavity (spinal cord)**

Heart and lungs

Stomach and liver

Kidneys and bladder

2. TRUE OR FALSE

The brain is located in the ventral body cavity.

Answer: **False**

Explanation shown if missed: The dorsal (back) cavity has the cranial cavity, which holds the BRAIN, and the spinal cavity, which holds the SPINAL CORD.

Thoracic cavity

1. MULTIPLE CHOICE

The heart sits in the ___ cavity, and each lung sits in a ___ cavity.

✓ **pericardial; pleural**

pleural; pericardial

cranial; spinal

abdominal; pelvic

2. TRUE OR FALSE

The thoracic cavity contains the heart and lungs.

Answer: **True**

Explanation shown if missed: The thoracic cavity (chest) holds the heart and lungs. The heart sits in the PERICARDIAL cavity; each lung sits in its own PLEURAL cavity.

Mediastinum

1. **MULTIPLE CHOICE**

The mediastinum is:

✓ **The central compartment of the thoracic cavity**

The space around each lung

The lining of the abdominopelvic cavity

Part of the dorsal body cavity

2. **SELECT ALL**

Select the structures found in the mediastinum.

✓ **Heart**

✓ **Esophagus**

✓ **Trachea**

✓ **Thymus**

Lungs

Kidneys

3. **TRUE OR FALSE**

The lungs are located inside the mediastinum.

Answer: **False**

Explanation shown if missed: The mediastinum is the central compartment of the chest, between the lungs. It holds the heart, great vessels, esophagus, trachea, phrenic and cardiac nerves, thoracic duct, thymus, and lymph nodes. The lungs are NOT in it.

Serous membranes

1. **MULTIPLE CHOICE**

Which serous membrane is associated with the lungs?

✓ **Pleurae**

Pericardium

Peritoneum

Mediastinum

2. **MULTIPLE CHOICE**

Which serous membrane is associated with the abdominopelvic cavity?

✓ **Peritoneum**

Pleurae

Pericardium

Diaphragm

3. ANALYZE THE CASE

Each time the heart beats, it rubs against the tissue around it. What reduces this friction?

✓ **Serous fluid**

Blood in the dural sinuses

Air in the lungs

The diaphragm

Explanation shown if missed: Serous membranes are thin linings: PERICARDIUM = heart, PLEURAE = lungs, PERITONEUM = abdominopelvic cavity. Their serous fluid lets organs move and expand without friction.

Parietal vs. visceral layers

1. MULTIPLE CHOICE

The visceral layer of a serous membrane:

✓ **Lines/covers the internal organ**

Lines the cavity/body wall

Is filled with air

Is part of the skin

2. ANALYZE THE CASE

The layer of pleura attached directly to the surface of the lung is the:

✓ **Visceral pleura**

Parietal pleura

Pericardium

Peritoneum

3. TRUE OR FALSE

The parietal layer lines the cavity or body wall.

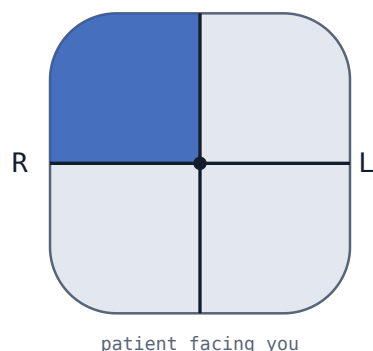
Answer: **True**

Explanation shown if missed: Picture pushing your fist into a balloon: the layer touching your fist is VISCERAL (covers the organ); the outer layer is PARIETAL (lines the cavity or body wall).

Abdominal regions & organs 4 topics

Abdominal quadrants

1. MULTIPLE CHOICE



Which quadrant is highlighted? (R = patient's right)

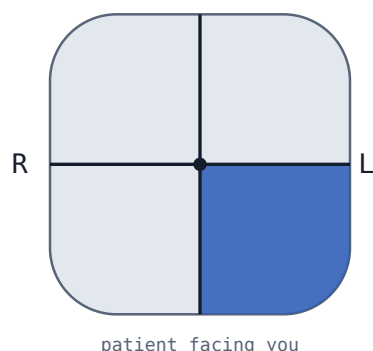
✓ **RUQ — Right Upper Quadrant**

LUQ — Left Upper Quadrant

RLQ — Right Lower Quadrant

LLQ — Left Lower Quadrant

2. MULTIPLE CHOICE



Which quadrant is highlighted? (R = patient's right)

✓ **LLQ — Left Lower Quadrant**

RLQ — Right Lower Quadrant

LUQ — Left Upper Quadrant

RUQ — Right Upper Quadrant

3. MULTIPLE CHOICE

What does LUQ stand for?

✓ **Left Upper Quadrant**

Lower Upper Quadrant

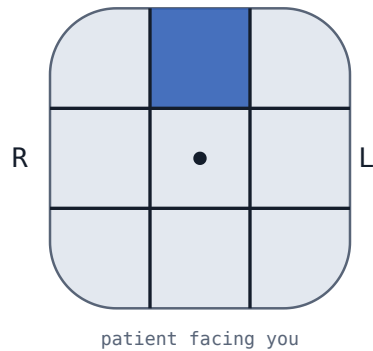
Lateral Upper Quadrant

Left Under Quadrant

Explanation shown if missed: The abdomen is split into 4 quadrants by a vertical and a horizontal line through the belly button: RUQ, LUQ, RLQ, LLQ. Right and left are always the PATIENT's right and left.

Epigastric, umbilical & hypogastric regions

1. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted?

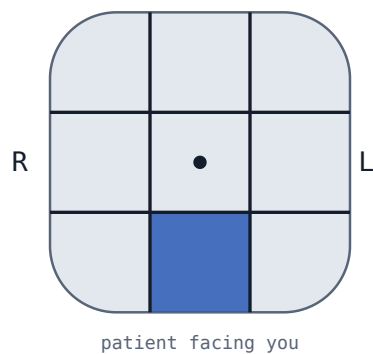
✓ **Epigastric**

Umbilical

Hypogastric

Right hypochondriac

2. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted?

✓ **Hypogastric**

Umbilical

Epigastric

Left iliac (inguinal)

3. MULTIPLE CHOICE

Which region is in the CENTER of the 9 abdominal regions?

✓ **Umbilical**

Epigastric

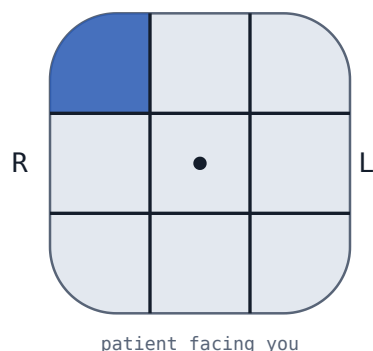
Hypogastric

Left lumbar

Explanation shown if missed: The middle column of the 9 regions, from top to bottom: EPIGASTRIC (above the stomach area), UMBILICAL (around the belly button), HYPOGASTRIC (below the belly button).

Hypochondriac, lumbar & iliac regions

1. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted? (R = patient's right)

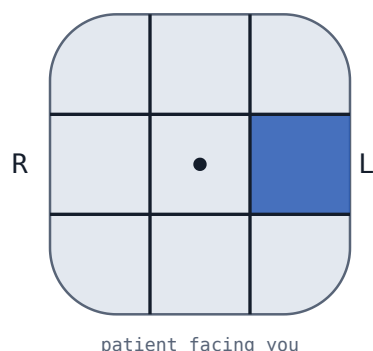
✓ **Right hypochondriac**

Left hypochondriac

Epigastric

Right lumbar

2. MULTIPLE CHOICE



Which of the 9 abdominal regions is highlighted? (R = patient's right)

✓ **Left lumbar**

Right lumbar

Umbilical

Left iliac (inguinal)

3. MULTIPLE CHOICE

Which regions make up the **BOTTOM** row of the 9 abdominal regions?

✓ **Right iliac, hypogastric, left iliac**

Right lumbar, umbilical, left lumbar

Right hypochondriac, epigastric, left hypochondriac

Right iliac, umbilical, left lumbar

Explanation shown if missed: The side columns, from top to bottom: HYPOCHONDRIAC (under the ribs), LUMBAR (beside the belly button), ILIAC/INGUINAL (near the hips and groin). Each has a right and a left.

Retroperitoneal organs (SAD PUCKER)

1.

What does retroperitoneal mean?

✓ **Situated behind the peritoneum**

Inside the pericardium

Above the diaphragm

Inside the cranial cavity

2.

Select the retroperitoneal organs (SAD PUCKER).

✓ **Kidneys**

✓ **Pancreas**

✓ **Abdominal aorta**

✓ **Ureters**

Brain

Lungs

3.

In the memory trick SAD PUCKER, what does the K stand for?

✓ **Kidneys**

Knees

Keratin

Kinetic energy

Explanation shown if missed: Retroperitoneal = BEHIND the peritoneum. Remember SAD PUCKER: Suprarenal glands, Abdominal aorta, Duodenum, Pancreas, Ureters, Colon (ascending/descending), Kidneys, Esophagus (inferior part), Rectum.

Clinical conditions & imaging 8 topics

Pleuritis & pleural effusion

1.

Pleuritis (pleurisy) is:

✓ **Inflammation of the pleurae**

Fluid in the peritoneal cavity

Compression of the heart

Inflammation of the peritoneum

2. ANALYZE THE CASE

A chest X-ray shows abnormal fluid collecting in the spaces around the lungs. This condition is:

✓ **Pleural effusion**

- Ascites
- Pericarditis
- Peritonitis

Explanation shown if missed: Pleuritis (pleurisy) = the pleurae around the lungs are INFLAMED.
Pleural effusion = extra FLUID builds up in the spaces around the lungs.

Pericarditis & cardiac tamponade

1. MULTIPLE CHOICE

Cardiac tamponade is:

✓ **Compression of the heart from blood or fluid in the pericardial sac**

- Inflammation of the pleurae
- Fluid buildup in the peritoneal cavity
- A tear in the diaphragm

2. TRUE OR FALSE

Pericarditis is inflammation of the pericardium.

Answer: **True**

3. ANALYZE THE CASE

After a car accident, blood collects around the heart inside its sac and squeezes it. This is:

✓ **Cardiac tamponade**

- Pleurisy
- Ascites
- Peritonitis

Explanation shown if missed: Pericarditis = the pericardium (sac around the heart) is inflamed.
Cardiac tamponade = blood or fluid fills the pericardial sac and SQUEEZES the heart so it can't pump well.

Peritonitis & ascites

1. MULTIPLE CHOICE

Ascites is:

✓ **Abnormal accumulation of fluid in the peritoneal cavity**

- Inflammation of the peritoneum
- Fluid around the lungs
- Inflammation of the heart sac

2. TRUE OR FALSE

Peritonitis is inflammation of the peritoneum.

Answer: **True**

Explanation shown if missed: Peritonitis = the peritoneum (lining of the abdominopelvic cavity) is INFLAMED. Ascites = extra FLUID builds up in the peritoneal cavity, making the belly swell.

X-ray & CT

1. MULTIPLE CHOICE

Which imaging method combines X-ray images from different angles to create cross-sectional images?

✓ **CT**

MRI

Ultrasound

PET

2. ANALYZE THE CASE

A child falls and the doctor wants a quick look at a possible broken arm bone. Which imaging is most useful for bone?

✓ **X-ray**

PET

Angiography

Nuclear scan

Explanation shown if missed: X-ray uses high-energy radiation that passes through the body; it is great for bones. CT combines many X-ray images from different angles to make cross-section "slices."

MRI

1. MULTIPLE CHOICE

Which imaging procedure uses radio waves?

✓ **MRI**

CT

X-ray

Angiography

2. ANALYZE THE CASE

A doctor needs a very detailed image of the soft tissues inside a knee. Which imaging method fits best?

✓ **MRI**

X-ray

Nuclear scan

Angiography

3. TRUE OR FALSE

MRI uses high-energy X-ray radiation.

Answer: **False**

Explanation shown if missed: MRI uses pulses of RADIO WAVES (with a strong magnet) and gives very detailed pictures of soft tissue. It is the imaging method in this chapter that uses radio waves.

Ultrasound

1. MULTIPLE CHOICE

Ultrasound creates images using:

✓ **High-frequency sound waves and echoes**

Pulses of radio waves

A radioactive tracer

X-rays from different angles

2. ANALYZE THE CASE

A pregnant patient has an imaging test that bounces sound waves off the baby to make a picture. Which test is this?

✓ **Ultrasound**

CT

PET

X-ray

Explanation shown if missed: Ultrasound sends high-frequency SOUND waves into the body and uses the echoes to create images, like a bat using echoes to "see."

Nuclear scan & PET

1. MULTIPLE CHOICE

A PET scan assesses:

✓ **Metabolic/physiological activity using a tracer**

Bone structure using sound waves

Blood vessels using a catheter only

Soft tissue using radio waves

2. TRUE OR FALSE

A nuclear scan uses a radioactive tracer.

Answer: **True**

Explanation shown if missed: A nuclear scan uses a radioactive TRACER (radiopharmaceutical). A PET scan also uses a tracer and shows how active tissues are (their metabolism).

Angiography

1. MULTIPLE CHOICE

Which procedure is used to visualize blood vessels and may involve contrast dye and catheterization?

✓ **Angiography**

Ultrasound

MRI

PET

2. ANALYZE THE CASE

A doctor threads a thin catheter into an artery and injects dye to look for a blocked vessel. This procedure is:

✓ **Angiography**

Nuclear scan

Ultrasound

X-ray of the bone

Explanation shown if missed: Angiography is used to see BLOOD VESSELS. It may use a contrast dye and a thin tube (catheter) placed in a vessel.